

ANSWER SCHEME TEST 2
CHEMISTRY II FSPC0034
22.03.2021

FOUNDATION STUDY
SEMESTER 3 YEAR 2020/2021

QUESTION 1

(a)

Before addition of HCl

$$[\text{NH}_3] = \frac{(100 \text{ cm}^3)(0.1 \text{ mol dm}^{-3})}{200 \text{ cm}^3} = 0.05 \text{ mol dm}^{-3}$$

$$[\text{NH}_4^+] = \frac{(100 \text{ cm}^3)(1.0 \text{ mol dm}^{-3})}{200 \text{ cm}^3} = 0.5 \text{ mol dm}^{-3}$$

$$\text{pOH} = \text{pKb} - \log \frac{[\text{NH}_3]}{[\text{NH}_4^+]}$$

$$= -\log (1.74 \times 10^{-5}) - \log (0.05/0.50)$$

$$= 5.76$$

(1m given if student use correct formula and correct answer)

$$\text{pH} = 14 - 5.76 = 8.24$$

QUESTION 1

b

	$\text{H}_3\text{O}^+(\text{aq})$	$+ \text{NH}_3(\text{aq})$	$\rightleftharpoons$	$\text{NH}_4^+(\text{aq})$	$+ \text{H}_2\text{O}(\text{l})$
Initial	0	$= (0.1 \times 100) / 1000$ $= 0.01 \text{ mol}$		$= (1.0 \times 100) / 1000$ $= 0.1 \text{ mol}$	
Change	$=$ $(0.1 \times 1.0) / 1000$ $= 1.0 \times 10^{-4} \text{ mol}$	$1.0 \times 10^{-4} \text{ mol}$		$1.0 \times 10^{-4} \text{ mol}$	
Equilibrium		0.0099 mol		0.1001	

$$\text{pOH} = \text{pK}_b - \log \frac{[\text{NH}_3]}{[\text{NH}_4^+]}$$

$$= -\log (1.74 \times 10^{-5}) - \log 0.0099 / 0.1001$$

$$= 4.76 + 1.005$$

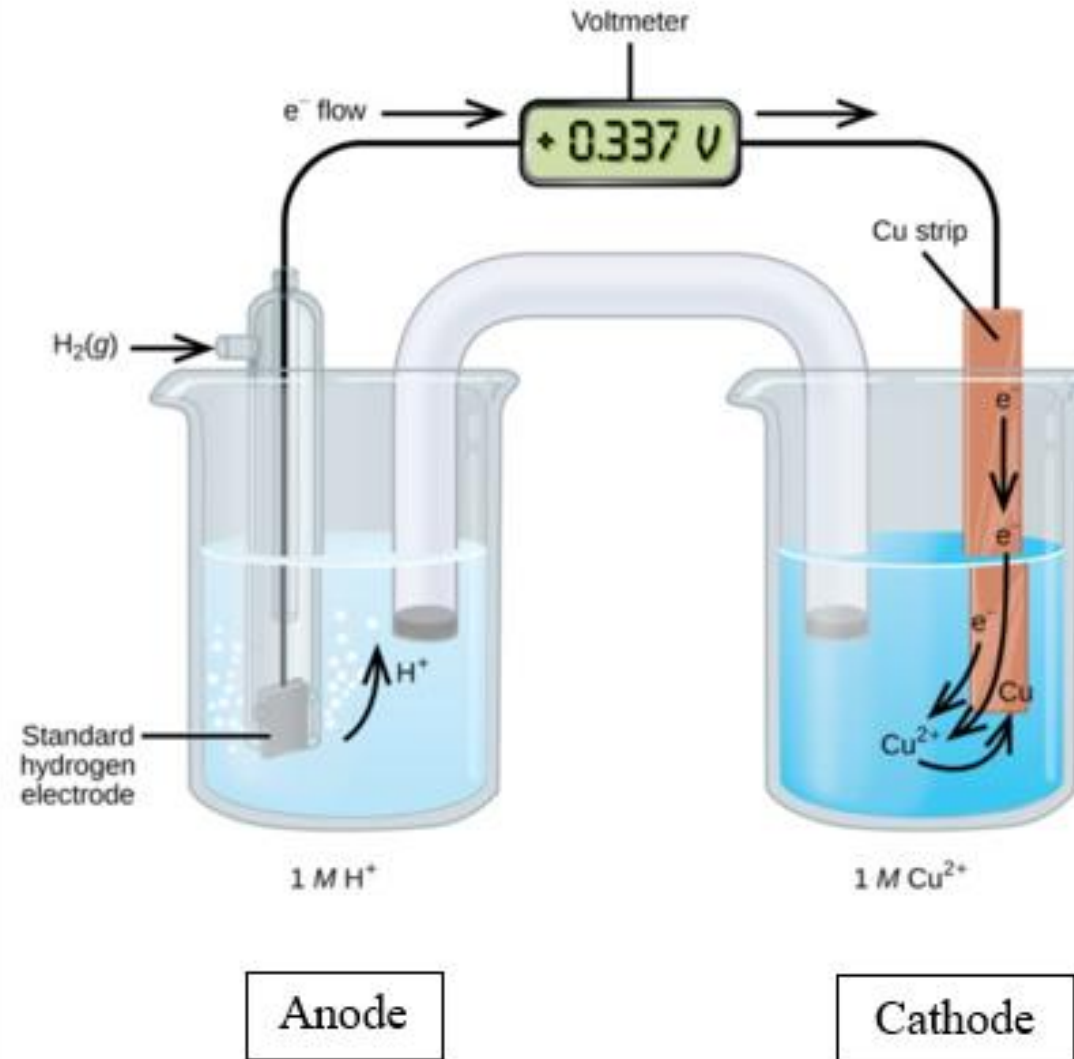
$$= 5.77$$

$$\text{pH} = 14 - 5.77 = 8.23$$

QUESTION 2

(a)

i



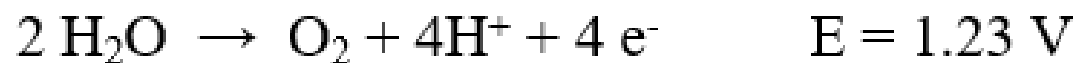
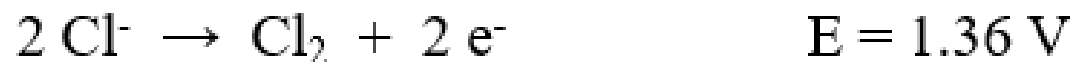
QUESTION 2

ii	Cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$
	Anode: $\text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$
iii	$\text{H}_2(\text{g}) + \text{Cu}^{2+}(\text{aq}) \rightarrow 2\text{H}^+ + \text{Cu}(\text{s})$
iv	$\text{Pt}(\text{s}) \text{H}_2(\text{g}) \text{H}^+(\text{aq}) \text{Cu}^{2+}(\text{aq}) \text{Cu}(\text{s})$
v	Cathode: Solid (copper) will be deposited on cathode. Anode: No observable change.

QUESTION 3

a

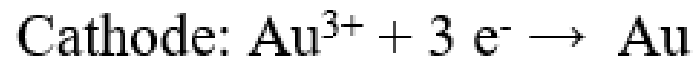
Anode:



Cathode:



Final product:



Overall Equation:



QUESTION 3

b

$$m = \frac{A \times I \times t}{Z \times F}$$

$$m = \frac{197 \text{ g/mol} (0.5 \text{ A}) (6\text{h} \times 60 \text{ mins} \times 60 \text{ s})}{(3 \text{ mol e})(96500 \frac{\text{cC}}{\text{mol e}})}$$

$$m = 7.35 \text{ g gold}$$

c

	Au	Sn
Charge	3	2
mole ratio	2	3

$$\begin{aligned} n_{\text{Au}} &= 10 \text{ g} / 197 \\ &= 0.051 \text{ mol} \end{aligned}$$

$$\begin{aligned} n_{\text{Sn}} &= 0.051 \text{ mol} \times 3/2 \\ &= 0.0765 \text{ mol} \end{aligned}$$

$$\begin{aligned} m_{\text{Sn}} &= 0.0765 \text{ mol} \times 118.71 \\ &= 9.08 \text{ g} \end{aligned}$$